SAHAYAK Presentation Guide

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Role: Live Demo, Testing and Conclusion
Suggested speaking time: 4 to 5 minutes, plus the demo

Your job in the presentation

You are the final speaker. Show the live website if it is working. Explain how to test the system fairly, tell the judges about its limits and future improvements, and finish the presentation. Prepare a backup in case the camera, internet or server fails.

Slide 12 - Introduce the demo

Say:

Thank you. I will show how someone can try the SAHAYAK prototype. I will explain what we should look for and how to judge its result. This project is still a prototype, so the demo can show both what it can do and where it may fail.

Before starting, make sure the audience can see the screen. Ask permission before using another person's image or recording them.

Live demo - step by step

1. Open the SAHAYAK website and choose Live Studio.
2. Check the internet and confirm that the backend is responding.
3. Allow camera access when the browser asks.
4. Put your hands clearly inside the camera view and wait for the hand-point overlay.
5. Make a sign that appears in the active model label list. Hold or move as required by the mode.
6. Point to the predicted label on the screen.
7. Try another supported sign and, when possible, one older sign.
8. If a result is wrong, say so honestly. Check the lighting and hand position once, then continue.

What to explain during the demo

Say:

The points drawn over the hand are landmarks found by the browser. The website changes them into numbers and sends the numbers to the backend. The trained model returns a predicted label, and the website displays it.

One or two correct results are only a demonstration. They cannot tell us the overall accuracy. To measure reliability, we need many repeated tests using examples that were kept separate from training.

Backup plan if the demo fails

- If the camera does not start, check browser permission and whether another app is using the camera.
- If the backend does not respond, say the prediction service is unavailable. Do not claim it made a prediction.
- If the hand points appear but the label is wrong, explain that hand tracking and sign prediction are separate stages.
- If internet is unavailable, explain that live backend prediction and some translation or speech features need a connection.
- If the demo still does not work, show the homepage or workflow diagram and talk through the steps.

You can say:

"The live result depends on camera detection and the hosted backend. We have a prototype, and this is a good example of why testing reliability matters. I will explain the workflow while we check the connection."

Slide 13 - How should we test it?

Say:

We should try each sign many times and record which results are right and wrong. Some test examples should come from people or recording sessions that were not used during training. We should test different lighting, camera angles, distances, hand positions and movement speeds.

Results for each sign can show which signs are hard for the model to tell apart. Response speed should be measured separately. A fast result can still be wrong.

Slide 14 - Limits and safe use

Say:

The current model recognizes only a limited set of trained labels. It does not understand all ISL sentences, face expressions, body movement or language rules. The hand detector can fail when the hands are hidden, outside the frame or in poor light.

SAHAYAK is not a guaranteed communication service or a replacement for an interpreter. Training examples should be collected with permission, stored carefully and kept separate from fair test examples.

Slide 15 - Future improvements

Say:

Future work includes collecting more varied examples with permission, adding signs only after training and testing, improving moving-sign recognition, reporting results for every label, and testing the website with people who use ISL. We also need clear privacy information and better error messages.

Final conclusion

Say:

To conclude, SAHAYAK shows how a camera, hand-point detection, machine learning and a website can work together. It is a starting point, not a finished universal translator. Useful assistive technology needs good data, fair testing, user feedback, privacy care and honest reporting. Thank you for listening. We welcome your questions.

Questions judges may ask

Q: What happens when the model is wrong?

A: We record the mistake and investigate the camera conditions and model result instead of hiding it.

Q: Does the full website work offline?

A: No. Live backend prediction and some translation or speech services need a connection.

Q: How can the model improve?

A: Collect more varied labelled examples, test on fresh data, inspect mistakes and retrain only after checking the candidate.

Q: What is its accuracy?

A: We should give a percentage only if we have a saved report from a fair test. Otherwise, we should say that the accuracy has not been properly measured yet.

Q: How do we protect privacy?

A: Ask for camera permission, explain what is sent or saved, get permission for training samples and avoid keeping unnecessary personal data.

Before the judges arrive

- Open the homepage and Live Studio.
- Check camera permission and internet access.
- Check that the backend is responding and the models are loaded.
- Check the current label list and test one or two supported signs.
- Make sure the hands are visible and the room is well lit.
- Keep a backup slide or workflow picture ready.
- Never invent accuracy numbers.

- If you do not know an answer, say you need to check it.